

Þ U N R E S D Æ G

EQUATIONS ON A BOARD

Equations on a board

Þunresdæg makes sound from **maps** — small equations turned a step at a time, the tent map and its relatives — and runs it through a **board** of effects you patch with cables.

Nothing is sampled. The equation oscillators make the numbers; two phasers, a bucket delay, seven pedals and a 4 × 4 matrix mixer take them, in any order, any output into any input — a box may even feed itself. Eight pads are not knobs but voltages: what they move is decided on the board.

It comes two ways: **ORIGIN** (the app's board, played with the pads) and **SYNTH** (keys play two equation oscillators, combined, through three effect slots). Each mode keeps its own sound and its own presets.

Too much to read? Skip it. In ORIGIN, open BOARD and hold RANDOM on the right a few times — each builds a new patch that reaches OUT. Close it with Done and play the pads.

In your DAW

The app, the AU (AUv3) and the VST3 are the same instrument. In your DAW you'll find **Þunresdæg** under *k.odk*. ORIGIN sounds on its own; SYNTH plays from MIDI notes. As an audio effect, **Þunresdæg FX** runs what comes in through the board (see page 11).

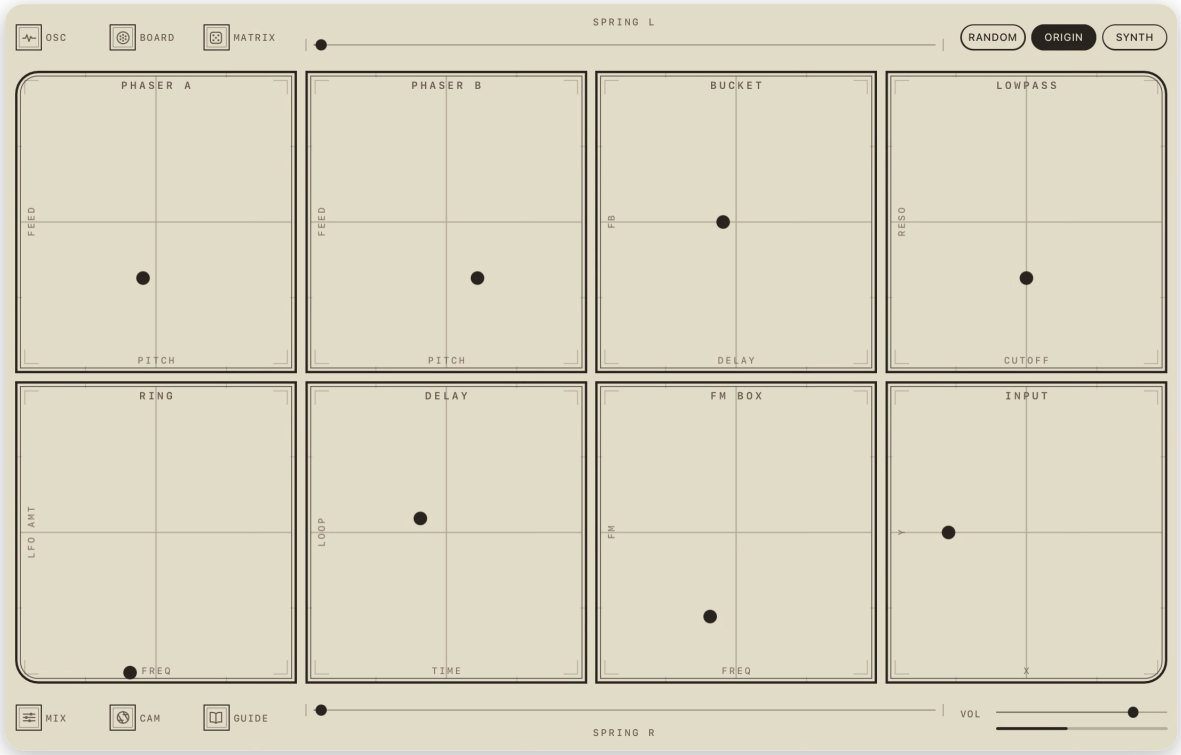
Installing

1. Open **Þunresdæg.pkg** and follow the installer. You can choose: **Standalone + AU**, and/or **VST3**.
2. Open **Þunresdæg** once from Applications — this registers the AUv3.
3. In your DAW, rescan plug-ins. Þunresdæg is under **k.odk**.

Standalone + AU	The Þunresdæg app in /Applications, with its Audio Unit (AUv3) inside — the instrument and the effect (Þunresdæg FX). Logic, GarageBand, Live and other AU hosts list them under *k.odk*.
VST3	In /Library/Audio/Plug-Ins/VST3 — Live, Bitwig, Reaper, Cubase …
The app	Plays on its own through the output chosen in the Mac's Sound settings.
Camera	Only for CAM. The first time, the Mac asks; if it was refused: System Settings › Privacy & Security › Camera (in a DAW, switch on the DAW). No picture is kept.

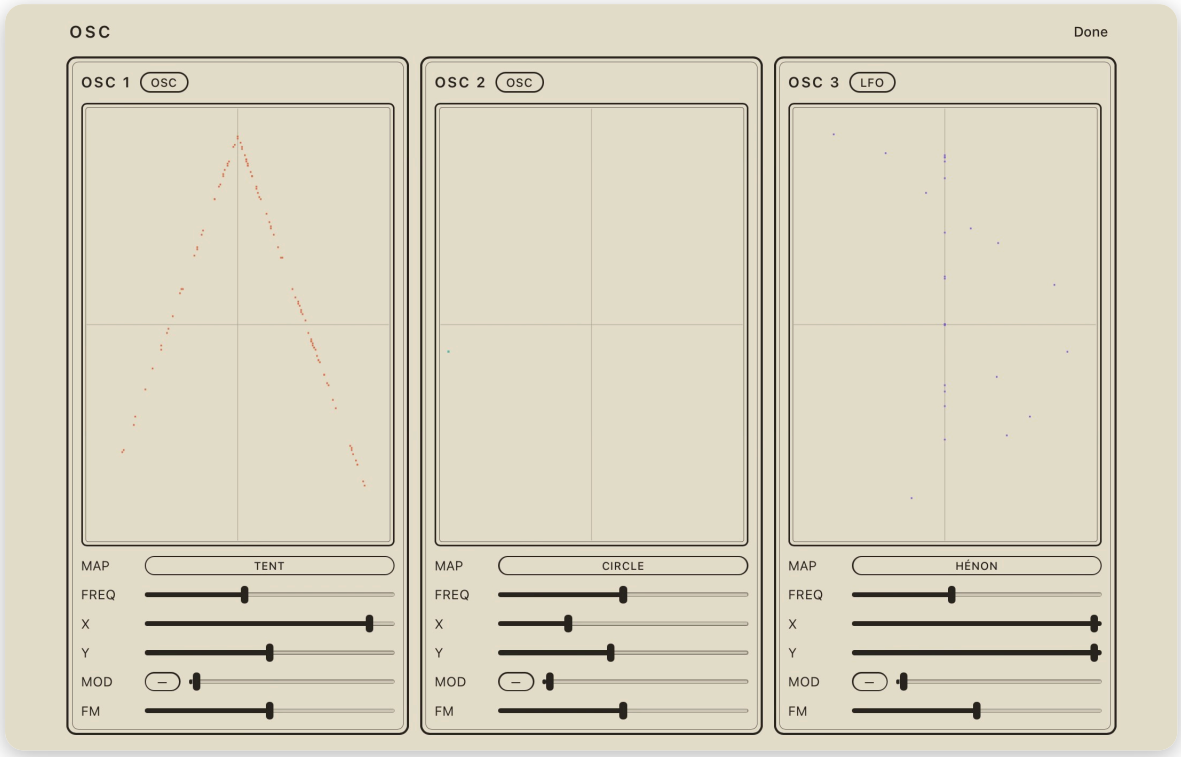
Requires macOS 12 or later · Apple silicon and Intel.

The two bars, and the pads



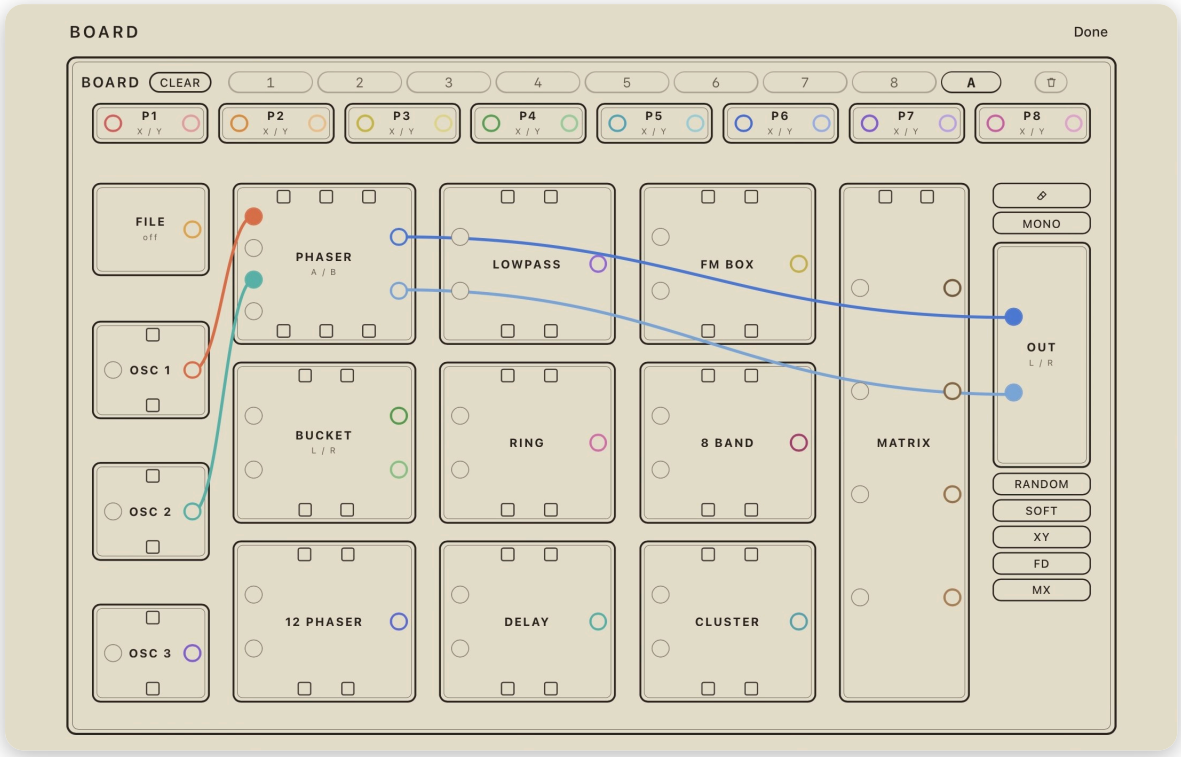
OSC · BOARD · MATRIX	The panels of ORIGIN: the oscillators, the board, the matrix and springs. **Done** or Esc closes them.
RANDOM · ORIGIN · SYNTH	RANDOM, held: a new sound in the mode on screen — the knobs go over slowly while it is held. Then the mode.
MIX · CAM · GUIDE	SETTINGS (mixer, file, camera, EQ, comp, limiter) · the camera on and off · a short guide.
The pads	Eight pads, four by two. Each gives two voltages, X and Y, to the board (P1 … P8). In the first patch they move PHASER A and B (PITCH, FEED), BUCKET (DELAY, FB), LOWPASS (CUTOFF, RESO), RING (FREQ, LFO AMT), DELAY (TIME, LOOP), FM BOX (FREQ, FM) and INPUT.
The sliders	ORIGIN: SPRING L above, SPRING R below — how much of each spring reverb. SYNTH: AMOUNT and LEVEL. VOL: the output, its meter under it.

Three equations, iterated



OSC · LFO	The chip by the name: OSC runs the map in the audio range, LFO in seconds — any map can be a voice or a slow hand.
MAP	TENT, LOGISTIC, SINE, CIRCLE, HÉNON, LORENZ, RÖSSLER, IKEDA, STANDARD.
FREQ	The clock that turns the map.
X · Y	The map's own two numbers: on the tent map X is the slope, on Hénon they are a and b — the same slider changes very different things.
MOD	Another oscillator's state (the chip: — 1 2 3 4) folded into this one, by the amount.
FM	How far the FREQ voltage on the board moves the clock.
The window	What the map is doing — a return map, a ring, an attractor — in the colour of its cable on the board.

Patch it with cables



Cables	Click an output (the ringed jack on a box's right), then an input: a cable. The same again takes it out. Round jacks are sound, square ones voltage.
Boxes	FILE and OSC 1 … 3 on the left; PHASER A/B, BUCKET, 12 PHASER, LOWPASS, RING, DELAY, FM BOX, 8 BAND, CLUSTER; then MATRIX and OUT L/R. Click a box: its own knobs open.
P1 … P8	The pads, X out on the left, Y on the right. Until a pad is wired it moves nothing.
Eraser · MONO	The eraser, then a jack: everything on it gone · one signal to both sides.
RANDOM SOFT XY FD MX	Held: new wires that always reach OUT · the wires stay, the knobs change · pads into the voltage jacks · wires back · everything through the matrix. None draws a random number: each steps a fixed sequence.
CLEAR 1 … 8 · A	CLEAR: every wire out. The presets: click to go back, hold to keep, the bin then a number to empty; the letter steps through eight banks.

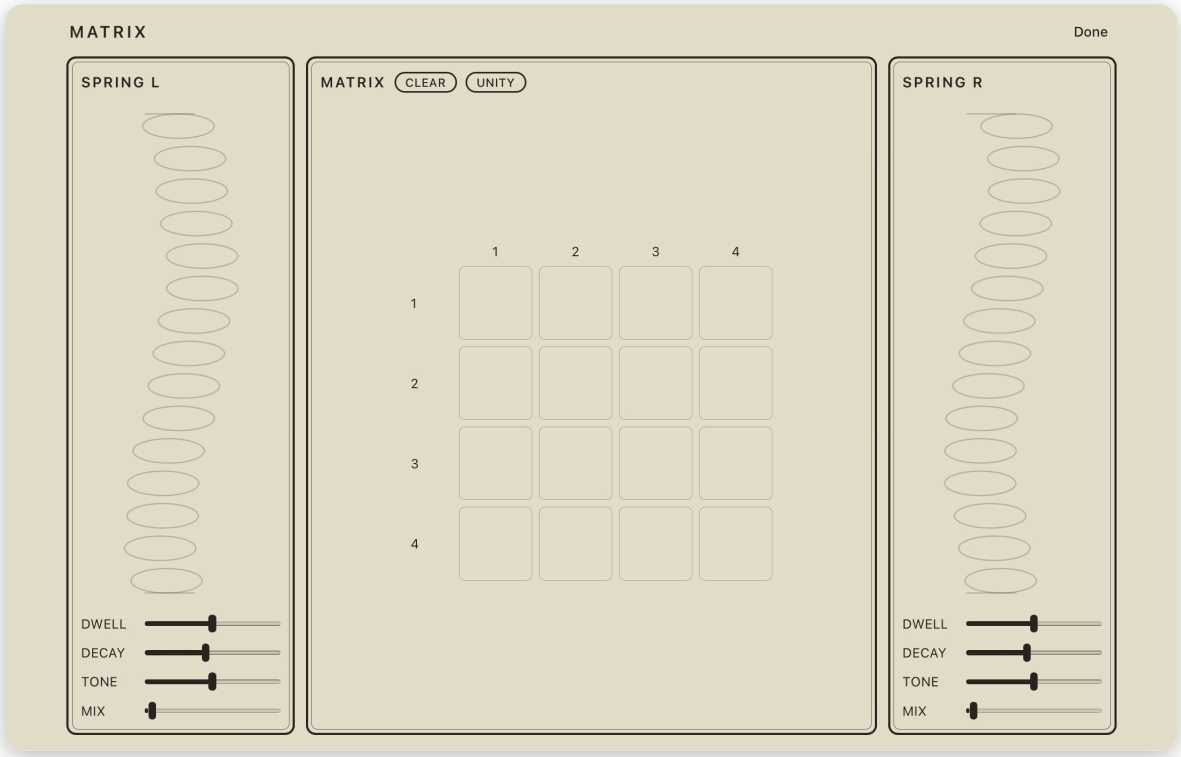
Feedback that sings

Click a box on the board and its own card opens; Done goes back to the board.

PHASER A · B	Six all-pass stages swept together, three taps (360, 720, 1080°). PITCH, FEED (from the tap chosen), LEVEL, CV. OUT: which tap goes to the board. Patch A into B and B into A and the stages double.
BUCKET	An analogue-style delay with no clock knob: DELAY is a voltage, and moving it bends the pitch. FILTER, FEED (middle: none, left: turned over), FLANGE. OUT A, B (twice as late) or FL.
12 PHASER	SWEEP, RESO, RATE, AMOUNT, DRIVE; 6 or 12 stages, LO or HI.
LOWPASS · RING	CUTOFF, RESO, ENV · FREQ, RATE, AMOUNT.
DELAY · FM BOX	TIME, FEEDBK, MIX · FREQ, WAVE, FM, ENV, MIX, SYNC.
8 BAND	Eight bands opened in turn by a PATTERN (1 … 8): RATE, ENV, MIX, DRIVE.
CLUSTER	CHORUS or FLANGE: TIME, FEEDBK, RATE, AMOUNT, MIX; the LFO SINE, TRI, SQR, SAW, RAMP, S&H.
CV · DRIVE	Every pedal has four voltage jacks: the first moves the main knob (CV is how far), the others add to the knob they are named for. DRIVE clips the input as a discrete stage does.
FILE	A sound file played into the board (MIX › OPEN). Mono, up to two minutes; the song keeps it with it.

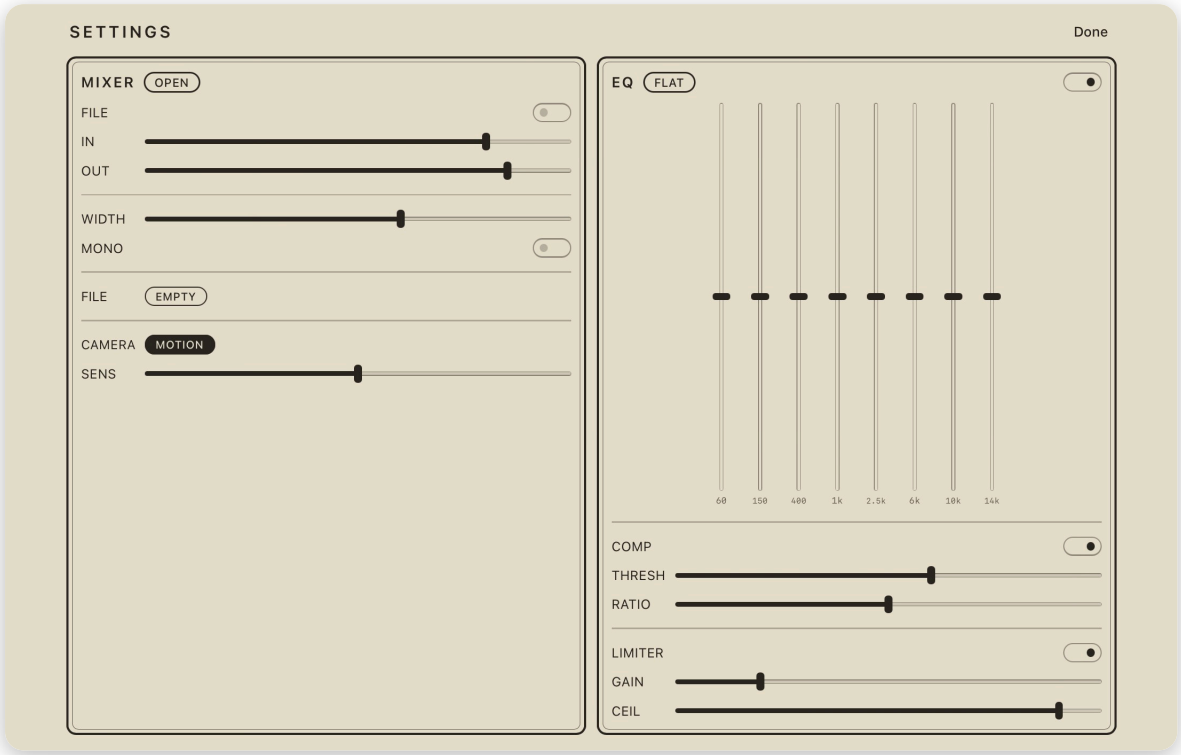
The pedals are written from how such circuits behave; they copy no product.

Where the paths cross



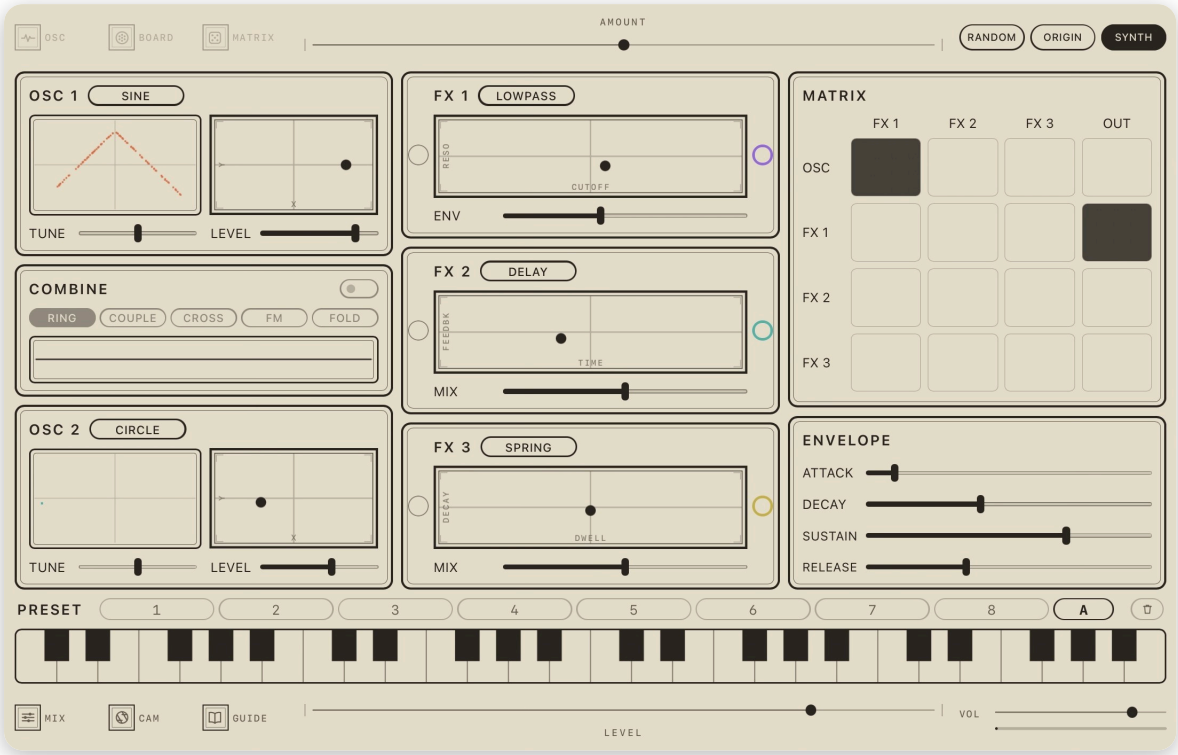
MATRIX	Four in, four out, one box on the board. A row is an input, a column an output: drag up or down in a square for the amount. UNITY sets the diagonal, CLEAR empties it.
SPRING L · R	Two spring reverbs, one each side of the output. DWELL (how hard they are driven), DECAY, TONE, MIX — the same MIX as the long sliders on the face. The drawing shakes with the sound.

The exit, and the camera



MIXER · OPEN	A sound file for the board's FILE box. Its switch plays it; IN is its level, EMPTY takes it out.
OUT · WIDTH · MONO	The master level · how wide · one signal to both sides.
CAMERA · SENS	With CAM on, each pad shows its piece of the picture and its point is drawn toward what moves there — or the brightest (BRIGHT), or the darkest (DARK). SENS: how much counts.
EQ	Eight bands, 60 Hz … 14 kHz; FLAT puts them back. Its switch.
COMP · LIMITER	THRESH, RATIO · make-up GAIN and CEIL. Each its switch. The order: level, EQ, compressor, limiter, out.

Two equations, played



OSC 1 · OSC 2	The chip chooses the map; the window draws it, the pad beside it holds X and Y. TUNE, LEVEL. Their clocks follow the key.
COMBINE	Its switch, then RING (one times the other), COUPLE (2 pulls 1), CROSS (each pulls the other), FM (2 moves 1's clock), FOLD (the sum folded). AMOUNT, the top slider, is how much; the window shows the two together.
FX 1 · 2 · 3	Each slot any effect (the chip); its two main knobs on the pad, the third on the slider below.
MATRIX	Rows: where it comes from (OSC, FX 1 … 3) · columns: where it goes (FX 1 … 3, OUT). Drag in a square. A slot may feed itself or another — it is rounded everywhere and cannot run away.
ENVELOPE	ATTACK, DECAY, SUSTAIN, RELEASE.

The slots, the keys

— (empty)	Nothing in the slot.
PHASER	PITCH · FEED · MIX
BUCKET	DELAY · FEED · MIX
12 PHASER	SWEEP · RESO · RATE
LOWPASS	CUTOFF · RESO · ENV
RING	FREQ · RATE · AMOUNT
DELAY	TIME · FEEDBK · MIX
FM BOX	FREQ · FM · MIX
8 BAND	RATE · ENV · MIX
CLUSTER	TIME · RATE · MIX
SPRING	DWELL · DECAY · MIX

The slot's knobs: X on the pad · Y on the pad · the slider.

Playing it

The keyboard	Four octaves on screen: click or drag. MIDI notes from the host play it too.
The Mac's keys	A W S E D F T G Y H U J K O L P ; — a row of white keys, the row above it black. Z and X: an octave down and up.
PRESET	1 … 8: click to go back, hold to keep · the letter: the next of eight banks · the bin, then a number: empty. SYNTH's presets are its own.
RANDOM	Held in SYNTH: new equations, slots and matrix.
LEVEL · VOL	The bottom slider: the synth's level · VOL: the output.

Þunresdæg FX, and a few ways in

On an audio track, **Þunresdæg FX** is the board alone: what comes in is the INPUT box (at first into both phasers), so the phasers, the bucket, the pedals and the matrix work on it. RANDOM keeps it in: every patch it builds starts from what comes in. A while after the sound stops coming in, it is quiet.

A first patch	BOARD › RANDOM a few times, then SOFT to shake the knobs while the cables stay.
Singing	Raise FEED on a phaser toward three quarters: several frequencies cross the edge at once and it turns to a dense noise.
Bend it	Wire a pad into the BUCKET's DELAY jack and move it slowly: what is in the line stretches.
Let the room play	CAM on, then wave a hand: the pads follow what moves.
SYNTH	COMBINE · FM, slot 1 LOWPASS, slot 2 SPRING, FX 1 → FX 2 → OUT in the matrix.
In a DAW	The host sees Volume and eight knobs for automation.

Credits

Þunresdæg — k.odk (Kei Odaka)
Support · keiodak.github.io/privacy/thunresdaeg-support.html · kodk.dev@gmail.com

Requires macOS 12 or later · Apple silicon and Intel.